The power of packages

Python has many advanced calculation capabilities that are used for data work and other scientific applications. These features make it possible to extend, enhance, and reuse parts of the code. To access these features, you can import them from libraries, packages, and modules. These features are not included in basic Python, so it's necessary to add them to your scripts. The additional functionality can save you time constructing functions and objects in your own work. Using these features can also help you obtain extra data types for analyzing data or building machine learning models.

Let's start with libraries.

A library, or package, broadly refers to a reusable collection of code. It also contains related modules and documentation. You'll often encounter the terms library and package used interchangeably. Commonly used libraries for data work are matplotlib and Seaborn. Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Seaborn is a data visualization library that's based on matplotlib. It provides a simpler interface for working with common plots and graphs.

The certificate program integrates two other commonly used libraries for data work, NumPy and pandas. NumPy, or numerical Python, is an essential library that contains multidimensional array and matrix data structures and functions to manipulate them.

This library is used for scientific computation. And pandas, which stands for Python data analysis, is a powerful library built on top of NumPy that's used to manipulate and analyze tabular data. There are many other popular Python libraries and packages for data professional work such as scikit-learn, statsmodels and others. Scikit-learn, a library, and statsmodels, a package, consists of functions data professionals can use to test the performance of statistical models. They're used across various scientific fields. Again, different practitioners across the field often conflate libraries and packages, so you may hear them refer to one, the other or both ways. Libraries and packages provide sets of modules that are essential for data professionals. Modules are accessed from within a package or a library. They are Python files that contain collections of functions and global variables. Global variables differ from other variables because these variables can be accessed from anywhere in a program or script. Modules are used to organize functions, classes, and other data in a structured way. Internally, modules are set up through separate files that contain these necessary classes and functions.
When you import a module, you are using pre-written code components. Each module is an executable file that can be added to your scripts. Commonly used modules for data professional work are math and random. Math provides access to mathematical functions, and random is used to generate random numbers. This is useful when selecting random elements from a list, shuffling elements randomly, or working with random sampling, which you'll explore in a later course. There are several ways to import modules depending on whether you want to use the whole package or just a single predefined function or feature. This adds functionality for carrying out specialized operations.

There's lots more to learn about libraries, packages and modules, so feel free to refer to the course resources for more information on installing these features and to continue growing your Python knowledge. But as a reminder, you don't have to install anything because everything you need to complete the different sections of the certificate program are already built into the notebooks you'll be using in Coursera. I'll introduce you to some libraries in the next video.

Library (or package)

Broadly refers to a reusable collection of code

Python libraries

- matplotlib
- Seaborn

matplotlib

A library for creating static, animated, and interactive visualizations in Python

Seaborn

A visualization library based on matplotlib that provides a simpler interface for working with common plots and graphs

Python libraries:

- NumPy
- pandas

NumPy

An essential library that contains multidimensional array and matrix data structures and functions to manipulate them

pandas

A powerful library built on top of NumPy that's used to manipulate and analyze tabular data

Module

A simple Python file containing a collection of functions and global variables

Global variables

Variables that can be accessed from anywhere in a program or script

Commonly used Python modules:

- Math
- Random